Explanations using Defeasible Rule Sets Literature Review

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Abstract

Explainable artificial intelligence requires not only that predictions be accurate but that the reasoning behind them be communicable, faithful, and formally grounded. This review examines the formal foundations of explanation generation using defeasible rule sets, motivated by the challenge of producing machine learning predictions that are genuinely explainable rather than merely interpretable. Two bodies of literature are developed in service of this goal. The first introduces Exception Closed Conjunctive Rule Sets (ECCRS), a symbolic rule-based classifier whose explanations are faithful by construction as it employs Most Specific Wins principle that produces traceable explanation traces without any external machinery. The second develops the KLM framework for defeasible reasoning and three forms of defeasible entailment within it: Rational Closure, System W, and Lexicographic Closure. Each is examined through its algorithmic procedure a nd illustrated using a running loan approval example. Rational Closure provides the conservative baseline for other defeasible entailments, System W and Lexicographical closure. The review establishes the formal connection between the two traditions.

1 Introduction

Machine Learning(ML) models achieve strong predictive performance but struggles to justify decisions in human-readable terms.They are trained to receive input and predict an output. As Machine Learning models are now deployed in high stake domains such medicine and finacial risk assessment [1], where decisions must be explained to people that they effect, the inability of humans to decipher how such predictions are made has motivated a growing concern around interpretability [1],[2], [3]. Interpretability refers to an extent to which humans can interpret a Machile Learning model's prediction [4], and has been proposed to as a means to engender trust in automated system [1]. However interpretability remains contested.Lipton [1] distinguishes between the transparency in a model's internal working and post-hoc explanations that tend to render output readable after the fact,noting that the two notions carries fundamentally different guarantees.Rudin [2] sharpenes this conecern into a direct argument:"a post-hoc explanation cannot be perfectly faithful to the model it describes, because if it were, it would simply be the model itself".In high-stakes applications, models should be inherently interpretable from the outset rather than explained afterward. Arrieta et al [3] support this position saying that opacity is specific to sub-symbolic methods, rule-based classifier and expert systems are readable by construction.

It is important to distuinguish interpretability, which concerns the model itself,from explainability,which concern the ability to produce artifacts, traces and justifications that makes makes a specific decision understandable [3]. While interpretability questions a model's construction, explainability asks whether concrete and communicable explanations can be produced for individual predictions.A model can be interpretable without producting useful explanations, a good explanation must do more than point to a transparent model, it must trace the specific reasoning path that led to a specific output.

Symbolic Rule-based classfier represent a natural candidate for inherently interpretable Machine Learning.By expressing decisions as explicit "if-then" rules over meaningful input features [5],they expose the reasoning behind each decision in the input structure in the model structure itself.However, not all rule-based formats produce faithful explanations.In ordered rule sets, predictions are order-dependent.The rules are tested sequentially and whichever matches the given input first resolve any kind of conflict and thus re-ordering the same rule sets can change the conclusion for the same instance [6]. An *instance* is a complete assignment of values to the input features describing a single entity e.g A student, a loan applicant or a patient.We say a rule apply to an instance when all of its rule are satiesfied by that instance.

In unordered rule sets ,rules are independent of their positions.Conflicts,where two rules with opposite labels both apply to the same instance, are resolved by external tie-breaking mechanism such as voting or preference scores, introducting computation that is separate from the rule and possibly absent from the explanation traces [7]. Darwiche et al [7] formalises this concern through the notion of *complete reason* which is the full characterazation of every property of the instance that could have caused the decision, and observe that explanations relying on this external mechinery are neccesarily incomplete with this standard. Exception Closed Conjunctive Rule Sets (ECCRS) are designed to meet this standard: conflicts between rules are resolved entirely through a structural specificity relationship encoded in the rule bodies themselves, producing explanation traces that are faithful, complete, and independent of any external machinery.

The structural design of ECCRS where most specific rules overide conclusion made by more general rule mirrors a pattern of reasoning studied formally in Logic for decades.Classical propositional logic cannot express this

pattern of overwritting conclusions.Its entailment relation is monotonic, meaning conclusions derived from a set of premise can never be retracted regardless of what new information is added [8]. Consider the classical example:*birds fly,birds have wings, penguins are bird, (yet)penguins do not fly* [9]. In classical logic this is a contradiction.No penguins can exists and classical entailment would permit any penguin based conclusions whatsoever. What we want to say is that birds *typically* fly , and penguins are exceptions but classical logic has not machinery to express *typicallity*. Defeasible reasoning was designed to address this address this limitation [10] ,providing a framework in which conclusions hold by default but may be retracted when more specific information become availible.

One of the central formal foundations for Defeasible Reasoning is the KLM framework introduced by Klaus, Lehmann and Magidor [9]. By extending propositional reasoning with defeasible conditionals of the form $\alpha \mid\sim \beta$, read as "typically if α,then I can conclude β if no new information says otherwise",KLM provided a systematic account of defeasible consequence and characterised classes of nonmonotonic inference relations through rationality postulates and corresponding semantic models [9]. We would then represent the Birds world with this condition, birds $\mid\sim$ fly . Within this tradition Lehmann and Magidor [8] later identified rational consequence as a more restricted class and proposed Rational Closure(RC) as a canonical form of entailment for finite knowledge base.

RC is intentionally cautious. It computes entailment by ranking formulas according to "exceptionality" and, for that reason, blocks the inheritance of general typical properties to exceptional subclasses when no further distinction is made [8]. This conservatism is illustracted by the penguin example,where a penguin is recognised as an exceptional bird and therefore refrain from concluding that penguins inherit all the default properties of birds.This is restrictive as RC would refrain from concluding that *penguings have wing*. Lehman [11] subsequently argued that this line of reasoning would fail in many defeasible setting and introduced Lexicographical closure as an alternative based on R Reiter's "A logic for default reasoning" paper [12] . Lexicographical Closure is less conservative, it draws conclusions about atypical entities unless the knowledge base contains specific evidence to the contrary [11]. Rather than ignoring all general statements about birds when reasoning about penguins, LC reasons more carefully about which specific statements are genuinely relevant to the exceptional case, and ignores only those. The result is that LC would conclude that penguins have wings since nothing in the knowledge base specifically contradicts this.

More recently, Komo and Beierie [13] introduced System W as a further development in the KLM framework.In some cases, RC is so cautious about atypical entities that it refuses to draw conclusions even when the atypicality is entirely unrelated to the conclusion being sought. System W addresses this by reasoning more carefully about which defaults are genuinely relevant to the exceptional case at hand, producing strictly more conclusions than Rational Closure while remaining more disciplined than Lexicographic Closure.

This review therefore serves two purposes that together motivate its structure. First, it introduces ECCRS and establishes the formal properties , exception closure and Most Specific Wins , that make its explanations faithful and complete. Second, and more centrally, it develops the formal background in defeasible reasoning that explanation generation from defeasible rule sets depends on. Section 2 covers ECCRS ,its syntax, semantics, exception closure property, and Most Specific Wins prediction rule. Section 3 goes into more details on the KLM framework and examines three forms of defeasible entailment mentioned above further: Rational Closure, System W, and Lexicographic Closure, each presented through its intuition, formal definition, and a worked example. Lexicographic Closure receives the most detailed treatment, as it is the entailment relation the workbench this project builds is aligned with.

2 Exception-Closed Conjuctive Rule Set

2.1 Motivation

Consider a bank's automated loan approval system. The bank has developed a lending policy governing when applicants should be approved or denied loans based on their financial profile. The relevant features are defined in Table 1.

Feature	Meaning
Stable	Applicant has stable employment
Income	Appicant meetgs minimum income threshold
Credit	Applicant has good credit history
Default	Applicant has a prior loan default
Fraud	Applicant has a fraud flag on record

Table 1: Bank loan policy features

The initial policy is simple. Stable employed applicants who meet income requirements are approved but if they have a prior default they are denied. Good credit alone is sufficient for approval but if they have a fraud flag they are denied.Mazime et al [14] introduced the Defeasible Horn Classifier with Exceptions (DHCE) that handles this policy so well. DHCE is a rule-based classifer that makes default-exceptions explicit by attaching each exception directly to its parent default rule. This applies the "normally…unless…" principle making decisions easier to trace without using any post-hoc explanations. The DHCE rule set for this policy might look like this:

Rule	Condition	Decision
r1	stable ∧ income	approve
r2	stable ∧ income ∧ default	deny
r3	credit	approve
r4	credit ∧ fraud	deny

Table 2: The loan approval rule

The classifer is tested with the following applicants.

1. Auggie , who has a stable employment and minimum income required (stable ∧ income) is approved for a loan. *(r1 applies)*
2. Toni also has stable employment and minimum income required and has a prior loan default (stable ∧ income ∧) and is denied of loan. *(r1 fires but r2 overrides).*
3. Dani has good credit and has fraud flag (Credit ∧ Fraud) and is denied of loan. *(r3 fires but r4 overrides).*

Each trace is short, complete, and faithful. However, DHCE imposes two constraints that limit its formal scope.

First,Exceptions are attached only one level deep, exceptions to exceptions of rules are not supported.This choice keeps every override local to its default and preserves a short, readable trace where each default appears next to its own exception [14]. If the bank's risk team revise the policy.They observe that applicants with a prior default who can provide a secured asset against the loan represent an acceptable risk and should be approved. A new feature secured could then be introduced.

Rule	Condition	Decision
r5	stable ∧ income ∧ default ∧ secured	approve

This is an entirely reasonable policy revision. Rule r5 is an exception to r2, which is itself an exception to r1. The chain reads naturally: *"stable income applicants are approved, unless they have a prior default, unless they can secure the loan.* Now consider an a fourth applicant, Dave.

4. Dave with stable employment, meets income requirements, has a prior default, but has secured the loan.

The correct decision under the revised policy is **approve**:r5 overrides r2, which had overridden r1.This three-level chain cannot be represented in DHCE. Rule r5 would have to be encoded as a separate independent default, losing its structural connection to r1 and r2, and with it the readable exception narrative.

Secondly,the priority machinery (of DHCE), while largely intrinsic, still would need to assign numeric priorities externally to handle ocnflicts between default , lower numbers fires first. The priority machinery is, in the language of Darwiche and Hirth [7], outside the complete reason for the decision , part of the computation but absent from the rule structure. These two limitations share a common root: conflict resolution in DHCE is only partially intrinsic. What is needed is a classifier in which conflict resolution is fully intrinsic at every level, exception chains are representable to arbitrary depth, and predictions are formally certifiable by an independent reasoner. Exception Closed Conjunctive Rule Sets (ECCRS) are designed to meet exactly these requirement.

2.2 Syntax and Semantics

ECCRS operates on boolean features.Each instance is represented by a complete feature, an assignement of exactly one signed literal, either a_i or $\neg a_i$ for every feature atom a_i in the domain. David's profile would be: $F_{David} = \text{Stable} \wedge \text{Income} \wedge \neg\ \text{Credit} \wedge \text{Default} \wedge \text{Secured} \wedge \neg\ \text{Fraud}$

A rule is a pair (B, l) where B is a **consistent conjuction** of signed literals(The rule body ,encoding the condition under which the rule apply) and $l \in \{0, 1\}$ is the class label (The prediction the rule make if the conditions are met). The rule policy translates directly:
r1: stable ∧ income ⇒ 1
r2: stable ∧ income ∧ default ⇒ 0
...
where label 1 means approve the loan and label 0 means deny the loan. A rule (B, l) is **applicable** to an instance x if the instance satiesfies every literal in the body in the body B, that is $x \models B$. Intuitively, a rule applies when all of its conditions are met by the instance being classified.

Not every rule in a ruleset is relevant to every instance. When classifying a loan applicant, only the rules whose conditions are actually satisfied by that applicant's profile should be considered. A rule that requires good credit history is simply irrelevant to an applicant who does not have good credit , it cannot fire ,it cannot contribute to the prediction, and it should not appear in the explanation. A set of applicable rule for a given instance is the pools of rules that are genuinely in play for that classification decision.
Given a ruleset $\mathcal{R}$,A complete feature pattern F and a rule $(B, l) \in \mathcal{R}$ is *applicable* to F if F satiesfies every literal in the rule body B, i.e $F \models B$. A set of applicable rules F is:

$$\mathcal{A}(F) = \{(B, l) \in \mathcal{R} : F \models B\}$$

For David, $\mathcal{A}(F_{David}) = \{r1, r2, r5\}$ These rule are a cnadidate for determining the David's classification.

2.3 Exception Closure

Two rule bodies B and C comparable if $B \subset C$ or $C \subset B$ one is a strict subset of the other, meaning one rule is strictly more specific. The bodies of $r1, r2$ and $r3$ form a strict inclusion chain: $B_{r1} \subset B_{r2} \subset B_{r3}$. This hierarchy is the structural relationship ECCRS uses to resolve conflicts with no external priority is needed.If two opposite-label rules are comparable, specificity determines the winner. If two opposite-label rules are compatible , they can both apply to

the same instance,but not comparable, no structural basis for resolution exists and external tie-breaking machinery would be required, placing the resolution outside the complete reason for the decision [7].

Exception closure is the constraint that prevents this situation from arising. A ruleset is *relaxed exception closed* on a training domain D, if for every pair of opposite label rules, (B, l) and $(C, 1-l)$, comparability is only required when both rules co-apply in some instance of the training Domain, D.This is enforceable from the D and has no guarantees about unseen feature combinations.The boolean feature space contains 2^n different pattern and the training data only covers a small fraction of this. A ruleset is *strictly exception closed* if comparability is required for every compatible opposite-label pair regardless of whether any training instance triggers both. This is enforced over the full Boolean feature space. Strict exception closure is a stronger requirement and is relevant to this project and ECCRS and defeasible entailment reasoning hold when the ruleset is *strictly exception closed*.

2.4 Most Specific Wins

Given a complete feature pattern F, the Most Specific Wins principle identifies the inclusion-maximal applicable rules, which are those not strictly overridden by any more specific applicable rule:

$\mathcal{M}(F) := \{(B, l) \in \mathcal{A}(F) : \nexists (C, l') \in \mathcal{A}(F) \text{ with } B \subset C\}$

The MSW prediction is then:

$$\mathrm{MSF}_{\mathcal{R}}(F) = \begin{cases} l & \text{If } \mathcal{M}(F) \neq \emptyset \text{ and all rules in} \\ & \mathcal{M}(F) \text{ have label } l \\ \\ \bot & \text{If } \mathcal{M}(F) = \emptyset \end{cases}$$

Under strict exception closure, $\mathcal{M}(F)$ is guaranteed to contain only rules of the same label on every covered pattern,the mixed-label case cannot arise.

The argument is direct :Suppose we have two rules (B, l) and $(C, 1-l)$ both belonging to $\mathcal{M}(F)$, both are inclusion-maximal and have opposite labels.They both apply to F, so $F \models B$ and $F \models C$,meaning B and C are compatible.By strict exception closure , compatible opposite-label rules must be comparable, either $B \subset C$ or $C \subset B$. But if $B \subset C$ then (B, l) is overidden by $(C, 1-l)$ and cannot be in $\mathcal{M}(F)$, a contradiction. Symmetrically for $C \subset B$.Therefore no two rules in $\mathcal{M}(F)$ can have opposite labels under strict exception closure.

When no rule applies, MSW returns $\bot$, meaning the classifier abstains rather than forcing a prediction.

For David, the inclusion chain $B_{r1} \subset B_{r2} \subset B_{r3}$. $r1$ is overrided by $r2$ which in turn is overriden by $r5$. Therefore $\mathcal{M}\left(F_{F_{David}}\right) = \{r5\}$, giving prediction approve

ECCRS is shown here to be able to reach conclusions that are traceable.There are other already established frameworks that would produce the same result as ECCR, therefore a questions one may ask is whether this newly established rule set frameowork align with them in some way.

3 Defeasible Reasoning

The introduction has established the KLM Framework as the formal foundation for Defeasible reasoning and identified Rational Closure , System W and Lexicographical Closure as three distinct entailment relations within it.We have not yet developed the precise relationship between these systems and the algorithmic procedures that defines them.

Formally these three systems are not independent of each other, they stand in a relationship of stritly increasing strength. Rational closure is the most conservatitive as it endorse the fewest conclusions and serve as a lower bound that any reasonable defeasible reasoner should meet [8]. System W and Lexicographical Closure strictly extends RC, every conclusion that RC draws, they also draws but draws additional conclusions in cases where RC is being overly cautious. LC is stronger still as it draws most conclusions of the three by preserving the maximum amount of knowledge when dealing exceptional cases [15].

This ordering matters for explanations. The conclusions a system endorses determine the explanations it can produce, a more conservative system produces fewer conclusions and therefore narrower explanations, while a more permissive system produces richer explanations that reflect more of the available knowledge. All three systems derive their conclusions from the same underlying computational foundation: a ranking of knowledge base statements by exceptionality, computed by the BaseRank algorithm. What differs between them is how they use that ranking when answering queries. Understanding these algorithmic differences is essential to understanding what kind of explanation each system can produce , which is the central concern of this review.

3.1 Defeasible Knowledge base and Materialization

A defeasible Knowledge is simply a collection of defeasible conditions encoding what typically holds in a domain.Before any entailment algorithm can rank or reason with this conditions, it needs a way to identify which conditions describe exceptional situations, those whose antecedents conflict with the typical behaviour described by the rest of the knowledge base. This identification is done through the materialisation of the knowledge base.

Formally, A defeasible Knowledge base is $\mathcal{K}$ is a finite set of defeasible conditions.Its materialisation , $\vec{\mathcal{K}}$ replaces every defeasible condition with its classical material counterpart [16],

$$\vec{\mathcal{K}} := \{\alpha \rightarrow \beta : \alpha \mathrel{|\!\sim} \beta \in \mathcal{K}\}$$

A conditional $\alpha \mathrel{|\!\sim} \beta$ is exceptional with respect to $\mathcal{K}$ if:

$$\vec{\mathcal{K}} \models \neg\alpha$$

That is,α conflicts with the typical behaviour of the full knowledge base [16]. The full loan knowledge base is:

$$\mathcal{K} = \begin{cases} \text{Stable} \wedge \text{Income} \mid\sim \text{Approve} \\ \text{Stable} \wedge \text{Income} \wedge \text{Default} \mid\sim \neg\ \text{Approve} \\ \text{Stable} \wedge \text{Income} \wedge \text{Default} \wedge \text{Secured} \mid\sim \text{Approve} \\ \text{Credit} \mid\sim \text{Approve} \\ \text{Credit} \wedge \text{Fraud} \mid\sim \neg\ \text{Approve} \end{cases}$$

Under materialisation, rule 2 , rule 3 and rule 5 are exceptional. They describes an atypical situations that conflicts with the typical behaviour of the knowledge base. Exceptionality is the key concept that all three entailment algorithms use to rank conditionals, and it is defined from the materialisation alone.

3.2 Rational Closure

Rational closure ranks the conditionals in the knowledge base by how exceptional they are, and when answering a query about an exceptional, it ignores those conditionals that describes typical behaviours that atypical behaviour violates and reasons from a more specific knowledge base.

3.2.1 BaseRank algorithm

RC uses BaseRank [17] to rank the conditionals in the knowledge base by their degree of exceptionality.It does this interatively: it starts with the full materialisation,$E_0 = \vec{\mathcal{K}}$ and repeatedly identifies which conditionals are exceptional, E_i with respect to the remaining ones E_{i-1}, peeling away the most general layer at each step until nothing changes. A conditional's rank is the last layer in which it appears before being removed, the higher the rank, the more exceptional the conditional.

Given as input , the knowledge base $\mathcal{K}$, BaseRank computes a ranking as follows: Given a knowledge base K with BaseRank ranking and a query $\alpha \mid\sim \beta$:

Algorithm 1: BaseRank

$i := 0$
$E_0 := \overline{K}$
while $E_{i-1} \neq E_i$ **do**
  $E_{i+1} := \{\alpha \rightarrow \beta \in E_i \mid E_i \models \neg\alpha\}$
  $R_i := E_i$
  E_{i+1}
  $i := i + 1$
end
$R_\infty := E_{i-1}$
if $E_{i-1} = \emptyset$ **then**
  $n := i - 1$
end
else
  $n := i$
end
return $(R_0, ..., R_{n-1}, R_\infty, n)$

This algorithm outputs an ordered tuple $(R_0, ..., R_{n-1}, R_\infty, n)$ which we use to define the Base rank.

The base rank, $br_{\mathcal{K}}$, of of a conditional $\alpha \mid\sim \beta$ is :

$$br_{\not\in}(\alpha \mid\sim \beta) := \begin{cases} i & \text{if } \alpha \rightarrow \beta \in E_i \setminus E_{i+1} \\ \infty & \text{if } \alpha \rightarrow \beta \in E_n \end{cases}$$

Conditions asssigned to rank ∞ are maximally exceptional. APplying the BaseRank to the first 3 rules in our loan knowledge base , we would get that:

$$E_0 = \vec{\mathcal{K}}$$
$$E_1 = \{\text{Stable} \wedge \text{Income} \wedge \text{Default} \mid\sim \neg\ \text{Approve},$$
$$\text{Stable} \wedge \text{Income} \wedge \text{Default} \wedge \text{Secured} \mid\sim \text{Approve}\}$$
$$E_2 = \{\text{Stable} \wedge \text{Income} \wedge \text{Default} \wedge \text{Secured} \mid\sim \text{Approve}\}$$

A ranking would follow from this. Crucially, this ranking reconstructs the same specificity ordering that ECCRS's MSW uses: more specific rule bodies correspond to higher-ranked conditionals.

3.2.2 The Rational Closure ALgorithm

Given the BaseRank ranking, answering a query $\alpha \rightarrow \beta$ works as follows : if α describes an atypical situation , one that is exceptional with respect to the current knowledge remove the lowest-ranked conditionals and try again. Keep removing until α until α is no longer exceptional, then check whether the remaining knowledge classically entails $\alpha \rightarrow \beta$ $\alpha \rightarrow \beta$. This simulates the defeasible reasoner's response to an atypical situation by ignore the general defaults that do not apply, reason from the specific knowledge that does.

We present the RationClosure algorithm which takes as input : A knowledge base K, and a defeasible implication $\alpha \mid\sim \beta$ and output: **true**, if $K \mid\sim \alpha \mid\sim \beta$, and **false** otherwise.

Algorithm 2: RationalClosure

$(R_0, ..., R_{n-1}, R_\infty, n) := \text{BaseRank}(\mathcal{K})$;
$i := 0$
$R := \cup_{j=0}^{n} R_j$;
while $R_\infty \cup R \models \neg\alpha \wedge R \neq \emptyset$ **do**
  $R := R$
  R_i;
  $i := i + 1$;
end
return $R_\infty \cup R \models \alpha \rightarrow \beta$;

3.3 System W

System W was introduced by Komo and Beierle [13] as a nonmonotonic inference relation for conditional knowledge bases. Its aim is to preserve reasonable default conclusion that may be lost in more conservative approaches, while still remaining grounded in a principled semantics. Komo et al presented System W as extending both system Z [18] and skeptical c-inference [19], rather than as a variant of Rational Closure. The key motivation is that exceptionality should be handled more carefully: being exceptional with respect to one property should not automatically block inheritance of other unrelated typical properties.

The central idea as outlined in the paper [13] is to compare worlds rather than simply rank formulas or discard whole layers of defaults. A world is preferred when, at the most important level where two worlds differ, it falsifies fewer conditionals. This gives System W a finer-grained notion of typicality than approaches that rely only on a single ranking of defaults. As a result, it can avoid undesirable behavior, where a subclass loses unrelated superclass properties merely because it is exceptional in one respect.

3.4 Lexicographical Closure

Rational Closure answers queries by discarding entire ranks of conditionals when an antecedent is exceptional, it removes all rank-0 conditionals, then all rank-1 conditionals, and so on, until the antecedent becomes consistent. This is a blunt instrument. Consider the classic penguin example with the following knowledge base:

$$\mathcal{K}_{birds} = \begin{cases} \text{Birds} \mid\sim \text{fly} \\ \text{Bird} \mid\sim \text{wings} \\ \text{Penguin} \mid\sim \text{birds} \\ \text{Penguin} \mid\sim \text{flies} \end{cases}$$

BaseRank assigns bird $\mid\sim$ fly and bird $\mid\sim$ wings to rank 0 . The conditional penguin $\mid\sim \neg$ fly is exceptional at rank 0 because treating bird $\mid\sim$ fly as strict creates a contradiction for penguins. When RC answers the query "penguin $\mid\sim$ wings", it finds that the penguin antecedent is exceptional and discards all rank-0 conditionals , including bird $\mid\sim$ wings. But bird $\mid\sim$ wings has absolutely nothing to do with whether penguins fly. It is discarded simply because it shares a rank with the conflicting conditional bird $\mid\sim$ fThe information that birds typically have wings is lost entirely, and RC refuses to conclude that penguins have wings.
Lehmann [11] identified this as a fundamental limitation of prototypical reasoning and argued for a different approach. Rather than discarding entire ranks, we should try to preserve as many conditionals as possible and discard only what is strictly necessary to make the exceptional situation consistent.This isn the core intuition of LC, while RC is maximally cautions with drawing conclusions , LC is maximally cautions about discarding knowledge [20].
The way LC achieves this is through a seriousness ordering over subsets of the knowledge base. The idea is that when we must choose which conditionals to ignore in order to handle an exceptional case, some choices are more serious and costly than others. Ignoring a highly specific conditional is more serious than ignoring a general one, because specific conditionals represent hard-won knowledge about exceptional cases. And among subsets of the same specificity level, ignoring more conditionals is more serious than ignoring fewer. LC selects the least serious subset to ignore the one that discards the minimum necessary knowledge at the minimum specificity level and draws conclusions from the maximum remaining knowledge [16].

Lehnman [11] suggest an ordering of subsets of our knowledge base , $\mathcal{K}$,based on their "degree of seriousness" using two criteria: the size of the set and the seriousness of the elements in a set. We define this notation formally. The seriousness ordering, $\underset{s}{<}$ is defined over subsets of the knowledge base. Given two subsets A and B of $\mathcal{K}.A$ is considered more serious than B if A contains conditionals with higher base rank , more specific and exceptional knowledge or if both contain conditionals of the same highest rank but A contains more of them. Intuitively, a subset that sacrifices specific knowledge is more serious than one that sacrifices only general knowledge, and a subset that sacrifices more conditionals is more serious than one that sacrifices fewer.
For each subset $A \subseteq \mathcal{K}$ may be associated with a $n+1$tuple of $< a_0, a_1, ..., a_{n+1} >$ where $n_i \in \mathbb{N}$ that is,

$$\begin{aligned} v(A) =< & |\{\alpha \mid\sim \beta \in A : br(\alpha) = \infty\}|, \\ & |\{\alpha \mid\sim \beta \in A : br(\alpha) = n\}, | \\ & \cdots \\ & |\{\alpha \mid\sim \beta \in A : br(\alpha) = 0\}| > \end{aligned}$$

This tuple counts, from left to right, how many conditionals in A have rank ∞ then rank n, down to rank 0.
The seriousness ordering is then the natural lexicographic ordering on these tuples:

$$A \underset{S}{<} B \iff v(A) \underset{lex}{<} v(B)$$

$\underset{LC}{<}$ is a modular partial order , it defines a ranked interpretation, $\mathcal{R}^{\mathcal{K}}_{LC}$ [16] , which we use to define a lexicographic ranked function, $r^{LC}_{\mathcal{K}}$ with respect to a knowledge base $\mathcal{K}$ as:

$$r^{LC}_{\mathcal{K}}(\alpha) := \min\{\mathcal{R}^{\mathcal{K}}_{LC}(v) : v \in \hat{\alpha}\}$$

3.4.1 DefeasibleEntailment

The DefeasibleEntailment algorithm [17] is a generalisation of RationalClosure that accepts a ranking function as a parameter. For Lexicographic Closure, the ranking function is the lexicographic rank, defined using the seriousness ordering above rather than the base rank used by RC. The key difference in execution is how the algorithm handles a rank when the antecedent is still exceptional after removing

lower ranks: instead of discarding the entire rank at once as RC does, LC weakens the rank by finding the least serious subset of conditionals within that rank whose removal makes the antecedent consistent.

The DefeasibleEntailment algorithm is presented below , taking as in input: A knowledge base K, a K-faithful rank function r, and a DI $\alpha \mid\sim \beta$

Algorithm 3: DefeasibleEntailment

$(R_0, ..., R_{n-1}, R_\infty, n) := \text{Rank}(K, r)$
$i := 0$
$R := \cup_{j=0}^{n} R_j$
while $\{R_\infty\} \cup R \vDash \neg\alpha \wedge R \neq \emptyset$ **do**
    $R := R$
    $\{R_i\}$
    $i := i + 1$
end
return $\{R_\infty\} \cup R \vDash \alpha \rightarrow \beta$

For the following Rank function, let$[\alpha] := \{\beta \mid \alpha \equiv \beta, \beta \in \mathcal{L}\}$, that is, all formulas logically equivalent to α.

Algorithm 4: Rank

Input: A knowledge base K and a K-faithful rank function r
Output: An ordered tuple $(R_0, ..., R_{n-1}, R_\infty, n)$
$R_\infty := \neg\Big(\bigvee_{r([\alpha])=\infty}[\alpha]\Big)$;
$n := \max\{i \in \mathbb{N} \mid \exists\alpha \in \mathcal{L} \text{ s.t. } r(\alpha) = i\}$;
if $n = 0$ **then**
    $R_0 := T$;
    $n := 1$;
end
else
    for $i := 0$ to $n - 1$ do
        $R_i :=_{\text{def}} \neg\Big(\bigvee_{r([\alpha])=i+1}[\alpha]\Big)$
    end for
end
return $(R_0, ..., R_{n-1}, R_\infty, n)$

The above algorithms are an extension of RC and can compute any rational defeasible entailment relation if given the appropriate input. Shifting focusing bank into , the rank function,$r^{LC}_{\mathcal{K}}$, which is both $\mathcal{K}$ −faithful and base ranking preserving [16], then it follows that lexicographic closure can be computes with the DefeasibleEntailment algorithm described above. When given the input, $\mathcal{K}$, $r^{LC}_{\mathcal{K}}$ and $\alpha \mid\sim \beta$, and if and only $r^{LC}_{\mathcal{K}}(\alpha) < r^{LC}_{\mathcal{K}}(\alpha)\alpha \wedge \neg\beta)$ or $r^{LC}_{\mathcal{K}} = \infty$ then the algorithm **DefeasibleEntailment** return true.

4 Related Work

The work reviewed in Sections 2 and 3 sits within a broader landscape of research connecting symbolic machine learning, defeasible reasoning, and explanation generation. This section briefly surveys related work that is not covered in depth in the main review.

Explanation frameworks for defeasible reasoning. Recent work has investigated explanation in KLM-style defeasible reasoning through the notion of weak justification. Everett, Morris, and Meyer extends the principle of weak justification to Relavant and Lexicographical closure [21]. Chama et al. extend this line of work to defeasible reasoning in description logics, focusing on Rational Closure for $\mathcal{ALC}$ [22]. From the classifier side, Rienstra studies explanations for symbolically encoded Boolean classifiers in the presence of non-monotonic background theories, arguing that ordinary sufficient reasons can become inaccurate or overly verbose when dependencies between features are ignored [23];

[7]. Taken together, these works share a common goal with the present review: producing explanations that are not merely readable, but formally grounded in the reasoning process that generated them.

Other symbolic defeasible classifiers. Beyond the ECCRS-based approach considered in this review, other symbolic methods are also relevant. Jaakkola et al. study short Boolean formulas (which Mazime et al compares to in her paper [14]) as explanations and classifiers, emphasizing compactness and interpretability [24]. Law, Russo, and Broda present ILASP as a framework for learning expressive answer set programs, including defaults, exceptions, and constraints [25]. These approaches differ in representational style and inference mechanism, but all contribute to the broader goal of making symbolic predictive models both accurate and explainable.

Extensions of the KLM framework. The defeasible reasoning literature extends well beyond the propositional setting considered in this review. Rational Closure was adapted to description logics by Casini and Straccia [15], and Lexicographic Closure was later extended to the same setting [26]. For a broader foundational treatment of KLM-style defeasible description logics, see Britz et al. [27]. C asini, Meyer, and Varzinczak then proposed a more general framework of rational defeasible entailment in which both Rational Closure and Lexicographic Closure appear, with Rational Closure as the most conservative member of the framework [17]. These extensions suggest natural directions for future work on ECCRS, especially in investigating whether the alignment between ECCRS and stronger defeasible entailment relations extends beyond the propositional Boolean feature setting.

5 conclusion

This review has traced a path from the practical problem of explainability in machine learning to the formal machinery of defeasible reasoning. Section 2 established that ECCRS achieves explainability by design with its default-exception structure that is intrinsic and its conflict resolution is structural. The closing question of Section 2 was whether these predictions are formally certifiable whether an independent reasoner operating on the same knowledge would reach the same conclusions. Section 3 developed the formal framework that answers this question. The KLM framework provides the logical foundation. Rational Closure provides the conservative baseline entailment relation, ranking knowledge base statements by exceptionality and reasoning from the most specific surviving knowledge. System W reasons more carefully about which defaults are genuinely relevant to an exceptional case. Lexicographic Closure goes further still by preserving the maximum possible knowledge when handling exceptional cases, producing richer conclusions and richer explanations by discarding only what is strictly necessary.What is needed is a way to tie the two traditions together.

Our work sits at the intersection. By translating an ECCRS ruleset into a KLM defeasible knowledge base and verifying that Lexicographic Closure reasoning produces the same predictions as Most Specific Wins. Several open questions remain. Whether the alignment extends to System W, whether it holds under relaxed rather than strict exception closure, and whether the framework generalises beyond binary classification of labelsare all natural directions for future work. This review provides the formal and conceptual foundations on which that work can build.

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